Smart Undergraduate Course Navigator
Intelligent Decision Support for Course Selection and Academic Performance Optimization at UMaT

End of Semester Project - Streamlit Implementation

University of Mines and Technology (UMaT)
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GitHub repository: https://github.com/dogahwisdom/smart-course-navigator

Abstract

This artefact documents a Streamlit-based decision support system that couples cohort analytics with supervised machine learning to forecast pass probabilities, recommend semester loads, and classify academic risk. Three classifiers-logistic regression, decision trees, and random forests-are trained on a realistic synthetic engineering dataset (≥1000 rows) spanning mining, geological, electrical, and mechanical programs. The best-performing model is serialized with joblib and invoked from a multi-page Streamlit experience built entirely in Python.

Introduction

Engineering undergraduates must balance accreditation requirements, laboratory intensity, and individual preparedness. The navigator operationalizes historical signals into individualized guidance.

Problem Statement

Students lack transparent, quantitative aids when sequencing demanding courses, while administrators need reproducible pipelines for refreshing analytics.

Objectives

- Deliver Streamlit dashboards for analytics, prediction, recommendations, and model governance.
- Train and compare three ML models with accuracy, precision, recall, and F1-score.
- Expose reasoning strings aligned with cohort behaviour and modeled probabilities.

Literature Review

Decision support systems integrate data, models, and interfaces for semi-structured decisions (Power, Sharda, & Burstein, 2015). Educational data mining applies classification to academic traces (Romero & Ventura, 2013). Responsible deployment foregrounds transparency and student agency (Williamson, 2018).

Methodology

Synthetic data emulate registrar-style fields; preprocessing scales numerics and one-hot encodes categoricals; class imbalance is mitigated via class_weight within estimators; evaluation uses a stratified hold-out split.

System Design

The Streamlit runtime orchestrates UI, calls utility modules for analytics, recommendations, and risk analysis, and loads the persisted sklearn pipeline from disk.

Figure 1. Streamlit-centric architecture

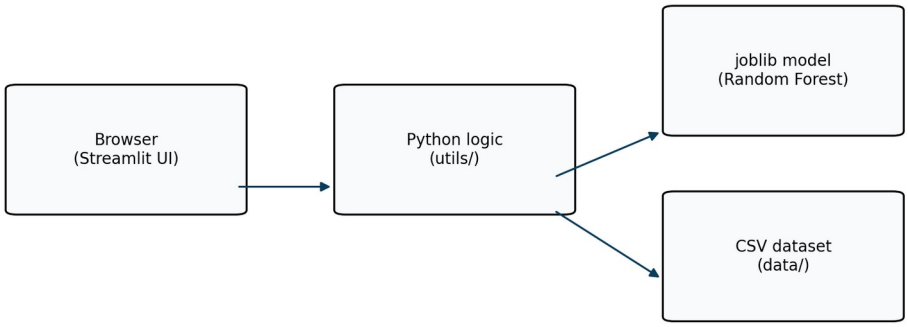
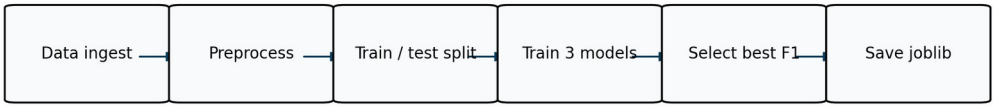


Figure 2. Machine learning workflow



Dataset Description

Fields include student_id, program, GPA, course identifiers, difficulty, attempts, attendance, credit load, grade, and pass/fail labels for supervised learning.

Machine Learning Approach

Selected model: random_forest. Highest F1-score on stratified hold-out test set (imbalance handled via class_weight).

Model	Accuracy	Precision	Recall	F1
logistic_regression	0.6507	0.851	0.639	0.7299
decision_tree	0.6027	0.7883	0.6318	0.7014
random_forest	0.712	0.8189	0.7834	0.8007

Figure 3. Model comparison (hold-out)

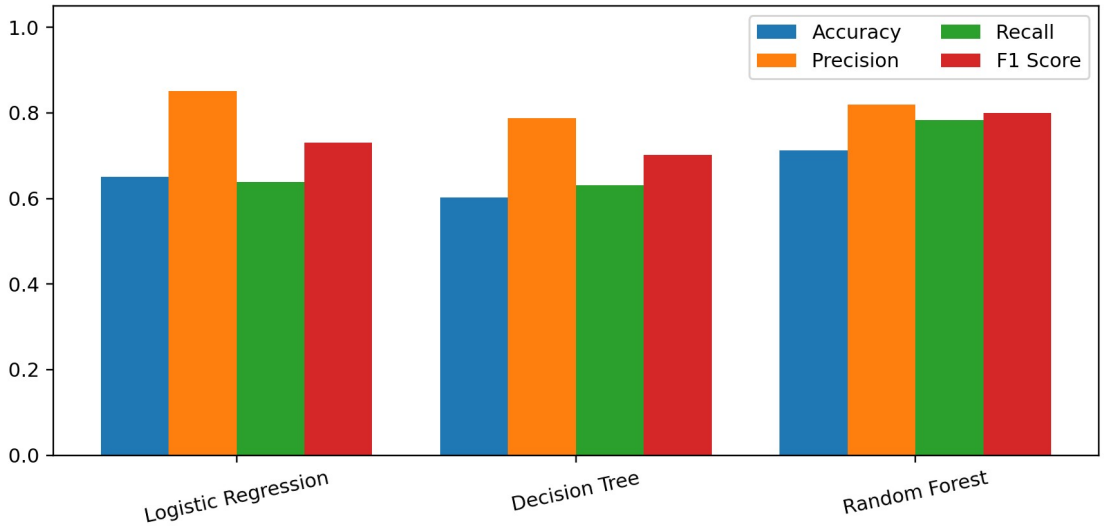
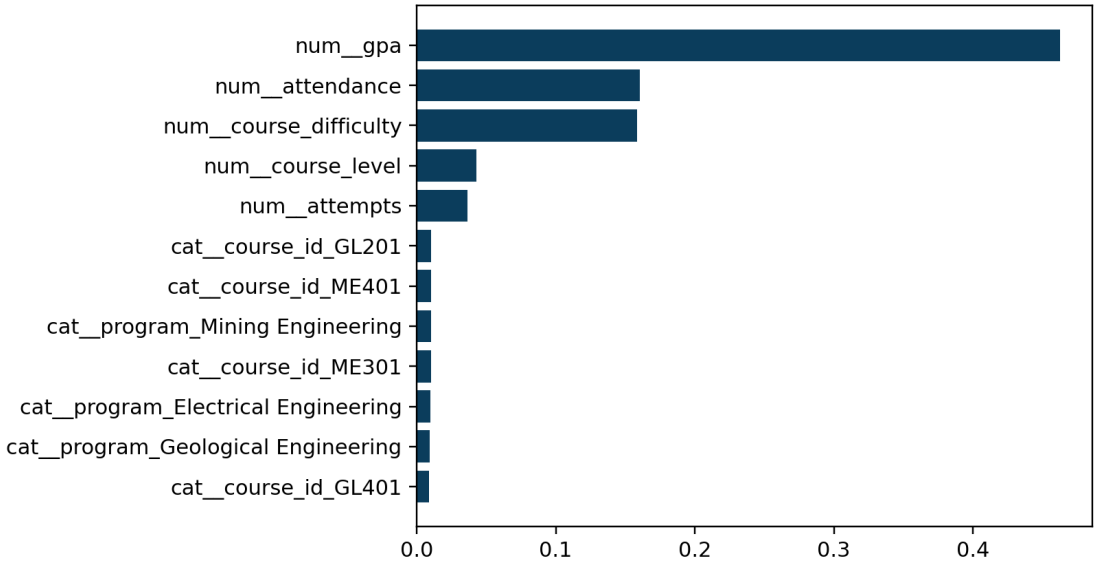


Figure 4. Random Forest feature importance (top encoded feat



System Implementation (Streamlit)

The implementation follows the prescribed structure: app.py for the home page, pages/ for modular views, utils/ for preprocessing and ML services, and notebooks/training.ipynb for reproducible training.

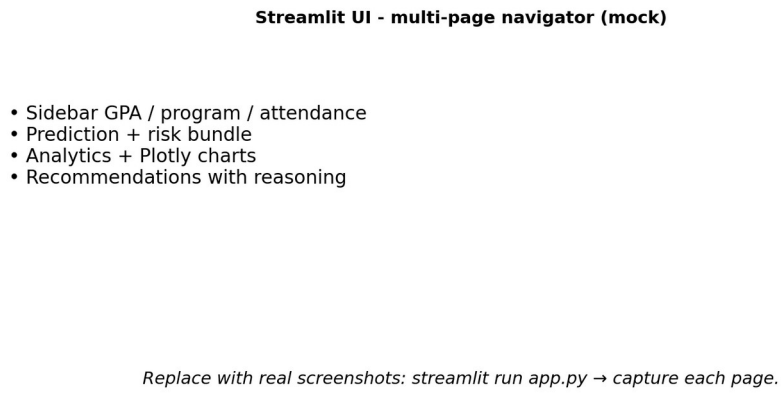


Figure: Streamlit user interface mock layout.

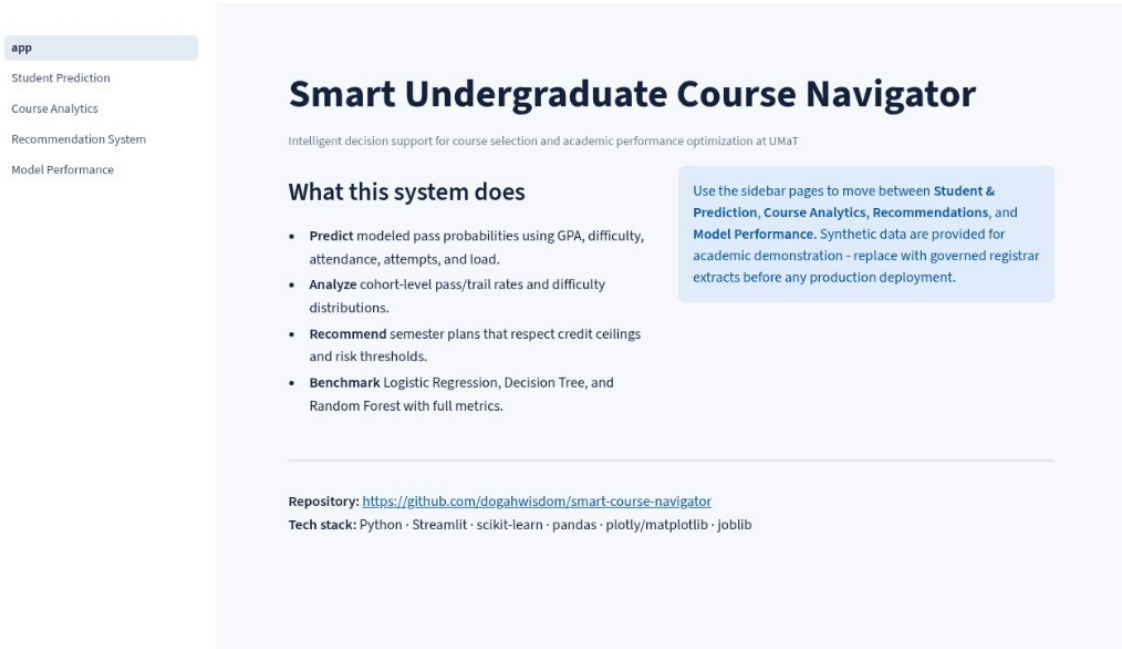


Figure: Home page with project overview and repository link.

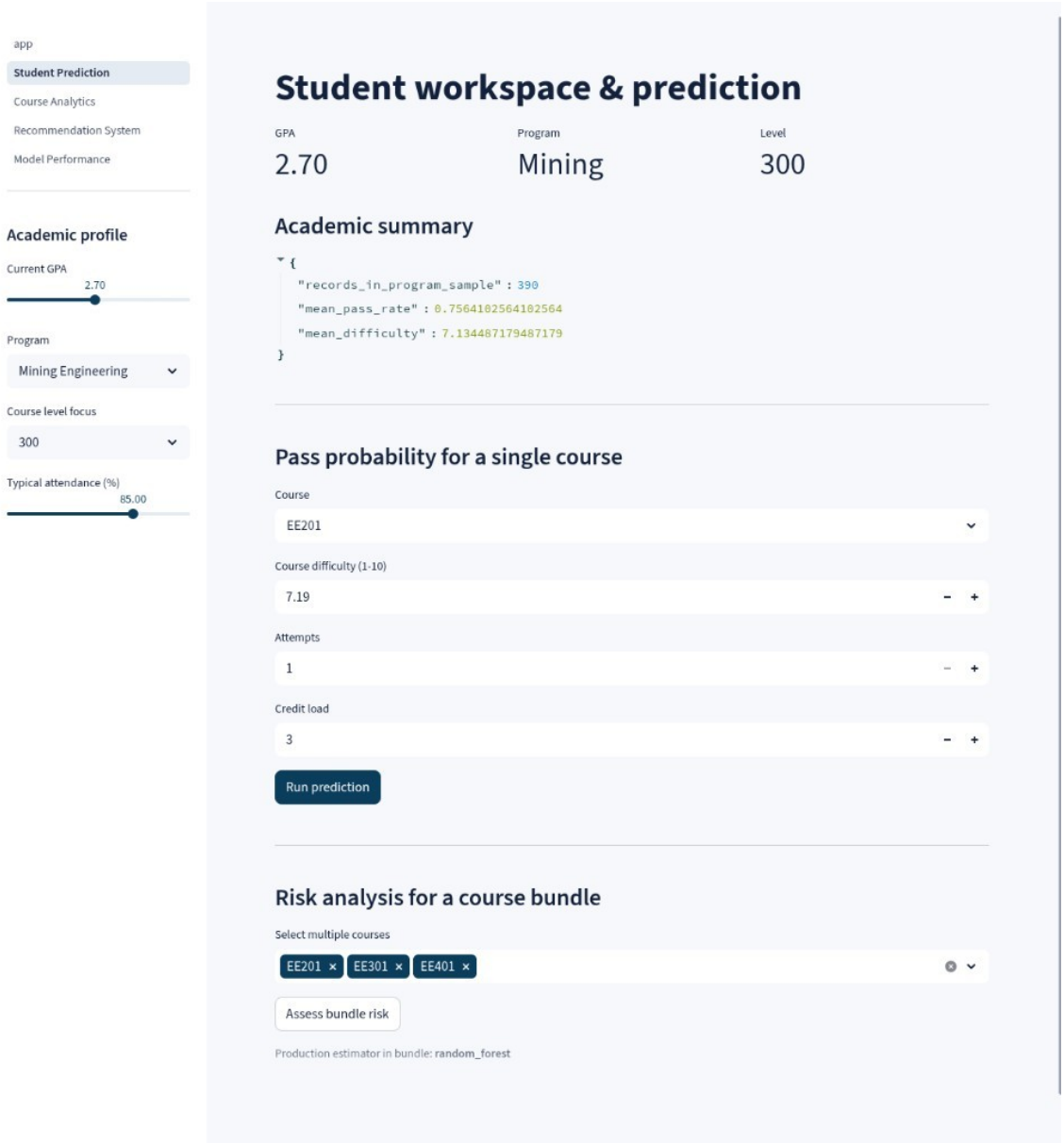


Figure: Student prediction page with profile inputs and risk analysis controls.

Figure: Course analytics page showing aggregated table and primary plots.

Figure: Extended analytics charts for difficulty and GPA-pass relationship.

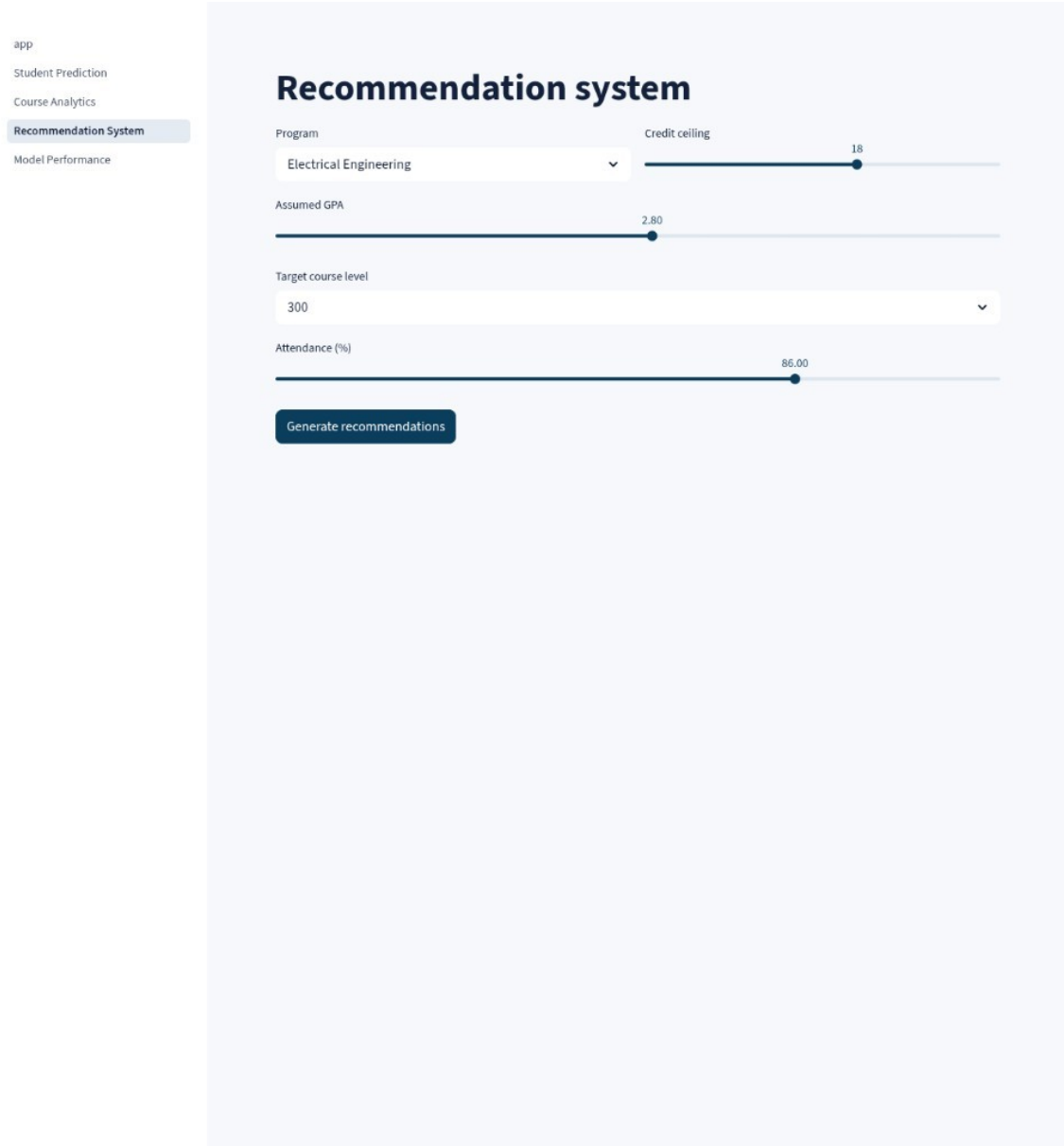


Figure: Recommendation page with program, GPA, level, and attendance controls.

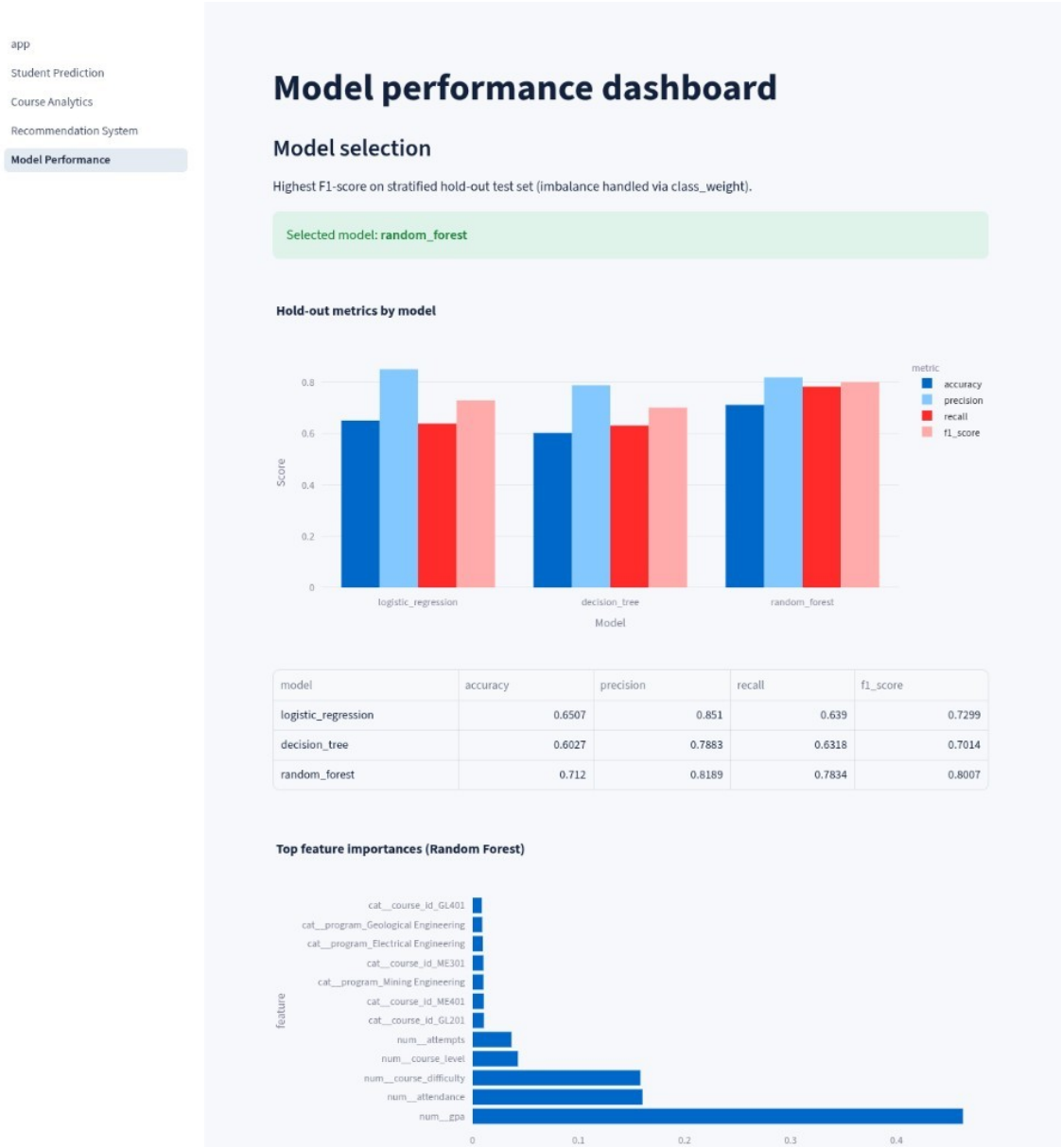


Figure: Model performance dashboard summary and model comparison chart.

Figure: Feature importance and interpretation guidance section.

Results and Evaluation

Quantitative metrics are summarized in the performance dashboard and embedded figures. Qualitative review confirms that reasoning strings map to cohort statistics and individualized probabilities.

Discussion

Synthetic data enable safe academic demonstration; production deployment requires governed extracts, bias audits, and integration with official advising workflows.

Conclusion

The Streamlit navigator satisfies the functional brief: prediction, analytics, recommendations, risk tiers, and model transparency within a single Python codebase.

Recommendations

Adopt institutional authentication, connect to relational databases, and schedule quarterly model retraining with drift monitoring.

References (APA-style samples)

- Power, D. J., Sharda, R., & Burstein, F. (2015). Decision support systems (3rd ed.). Wiley.
- Romero, C., & Ventura, S. (2013). Data mining in education. Wiley Interdisciplinary Reviews.
- Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

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Appendices

Appendix A: Directory tree (app.py, pages/, utils/, data/, models/). Appendix B: API-free architecture because Streamlit embeds server logic. Appendix C: evaluation_metrics.json stores raw metrics for audits.

Appendix D: Notebook evidence screenshots.

Model Comparison Metrics

model	accuracy	precision	recall	f1_score
logistic_regression	0.6507	0.851	0.639	0.7299
decision_tree	0.6027	0.7883	0.6318	0.7014
random_forest	0.712	0.8189	0.7834	0.8007

Figure: Notebook model metrics table export.

Top Feature Importances

feature	importance
num__gpa	0.462728
num__attendance	0.160313
num__course_difficulty	0.158306
num__course_level	0.042817
num__attempts	0.036641
cat__course_id_GL201	0.010598
cat__course_id_ME401	0.010476
cat__program_Mining Engineering	0.0102
cat__course_id_ME301	0.010186
cat__program_Electrical Engineering	0.00965
cat__program_Geological Engineering	0.009151
cat__course_id_GL401	0.008758

Figure: Notebook feature importance table export.